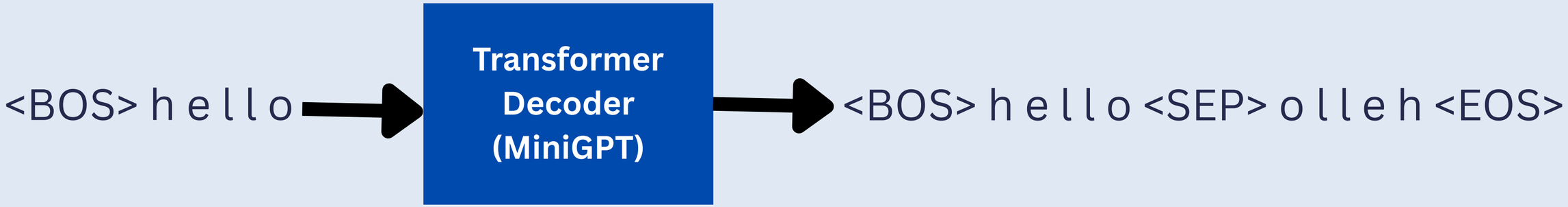
Mini Project - Lecture 1 Foundations of LLMs

Step 1 : Training GPT from scratch to reverse a sequence.

Given a sequence of English alphabets X = [x1 , x2 ,... xN]

train a transformer decoder to generate Y = [xN, xN-1, ..., x1]



Length of X = 2 - 6 characters.

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Step 2 : Fine-tune the trained model using reinforcement learning e.g. GRPO / PPO to make the model skip “b” (bomb) letter in the reversed output.

** Vibe coding allowed **

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Step 3 : Write a 6–8 page (A4) report that covers the following topics:

- GPT model configuration
- Preparation of training and evaluation datasets
- Accuracy/performance of the pretrained model
- Design of the RL / reward function
- Accuracy/performance of the fine-tuned model
- Discussion and analysis of the experimental findings

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Vibe coding (using AI tools such as ChatGPT, Copilot, or similar tools to assist with coding) is allowed. However, students must understand the generated code and clearly disclose any AI assistance used in their report.

Settings

```
# Transformer config
D_MODEL    = 128          # embedding dimension
N_HEADS    = 4            # number of attention heads
N_LAYERS   = 4            # number of transformer blocks
D_FF       = 128          # feed-forward hidden dimension
DROPOUT    = 0.1
```

Model with this setting can run without GPU !